

Project 1: Analyzing Soft-Margin SVM and Kernel SVM for Image Classification (Cougar-Face vs Cougar-Body).

In Lab 1, you have already implemented and experimented with a soft-margin SVM. In this project, you will apply SVM to a more challenging image-classification problem and investigate:

- how image representation affects linear separability;
- how the soft-margin parameter (C) changes the learned classifier;
- when a nonlinear kernel is useful;
- how kernel hyperparameters influence model complexity and generalization;
- what support vectors and misclassified samples reveal about the classification problem.

The emphasis is **not on writing large amounts of code**. Instead, marks will primarily be awarded for correct experimentation, appropriate evidence, and your ability to explain the observed behaviour using concepts from the lectures.

Before you start the assignment, please read through the following guidelines so that you understand the requirements. Follow the guidelines strictly to avoid unnecessary mark deductions:

1. Implement your codes with Python, MATLAB, or any programming languages you deem comfortable with.
2. For questions that require you to show your code, make sure your codes are **clean** and **easily readable** (add meaningful comments). Embed your code into your answer with screenshots.
3. Your submission should be **ONE** single PDF file with ALL the necessary code and results collated in a single file. Incorrect submission format will result in a **direct** 10 points (out of 100) deduction.
4. Submit your project on Canvas before the deadline: **23:59 (SGT), 4 Sep 2026**. There will be **10** submissions attempts restricted for everyone.
5. Policy on late submission: late submission will be deducted 10 points (out of 100) for every 24 hours.
6. This is an **individual** project, do **NOT** share your solutions with others, we have zero tolerance for **plagiarism**.
7. Please include an **AI Usage Statement** that states the AI tool used and how you have leveraged the AI tool at the **START** of your report.

Part 1: Soft-Margin Linear SVM: Implementation and Analysis (40%).

You will first investigate the behaviour of a soft-margin linear SVM on the cougar-face versus cougar-body classification problem. You may reuse and adapt your SVM implementation from Lab 1.

You will be provided with training and testing image folders for the two classes, zipped in “caltech_cougar.zip”. Please find the data in the Files tab. Each folder contains both training and testing images which you may split however you deem suitable. No features will be provided.

a. Data loading and feature construction (10%)

Convert every image into a fixed-length feature vector. For consistency, you should begin with the following baseline:

1. Resize all images to the same size;
2. Convert them to grayscale;
3. Flatten the image into a feature vector;
4. Normalize/standardize the resulting features.

You may modify this representation if you believe another simple representation is more appropriate, but complicated feature engineering is neither required nor expected. Clearly **report** the following information:

1. Training and test sample counts;
 2. Image preprocessing used;
 3. Feature dimension;
 4. Any feature normalization or standardization applied.
 5. **Explain in detail** why your representation is suitable, or maybe unsuitable, for distinguishing cougar-face from cougar-body.
-

b. Soft-margin linear SVM implementation (10%)

Apply the soft-margin SVM formulation introduced by minimizing the loss function:

$$L = \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_i \max(0, 1 - y_i(\mathbf{w}^T \mathbf{x}_i + b))$$

You may reuse your Lab 1 implementation rather than implementing the entire algorithm again. Remember to use the same training/test split throughout the project. Whether you reuse your implementation or not, your report should clearly **present**:

1. The implemented soft-margin SVM;
2. State the important training settings used;

3. **Explain in detail** how you determine the training settings that were selected.
-

c. Investigating the Effect of C (10%)

Instead of reporting only one final accuracy, systematically investigate how the behaviour of the classifier changes with parameter C . Use at least five values covering several orders of magnitude, for example: $C \in \{0.01, 0.1, 1, 10, 100\}$. For every C , **report**:

1. Training accuracy and test accuracy;
2. The actual margin $2/\|w\|$;
3. The number or proportion of training samples for which $y_i f(x_i) < 1$.

Present the results using an appropriate table and/or figure. Your analysis should address questions including but not limited to:

1. Does increasing C always improve training accuracy?
2. Does increasing C always improve test accuracy?
3. What happens to the margin as C increases?
4. What happens to the number of margin violations?
5. At what point, if any, does reducing training error stop improving generalization?
6. Are your observations consistent with the theoretical role of C discussed in the lecture?

You should use your experimental results as solid evidence.

d. Analyze the “Optimal” Trained Linear Classifier

Based on your implementation and actual experiments, Select the most reasonable linear SVM based on your experiments, and **report** the following:

1. Training accuracy and test accuracy;
2. The confusion matrix.

Then **examine** several representative test images, including both successful and failed predictions. Use these examples to **discuss in detail**:

1. Which class appears more difficult to classify;
2. What visual properties may cause failures;
3. Whether the problem appears approximately linearly separable using your chosen image representation;
4. Whether the main limitation appears to come from the classifier, the feature representation, or both.

Part 2: Kernel SVM: Comparative and Diagnostic Analysis (60%).

You will now investigate whether mapping the data implicitly into a different feature space using a kernel produces a more suitable classifier. For this part, you **may use `sklearn.svm.SVC`**. You are not expected to implement a numerical SVM optimizer from scratch.

a. Linear vs. Nonlinear Kernels in Kernel SVM (20%)

Train SVM classifiers using at least the following (you are encouraged to use others):

1. Linear kernel;
2. Polynomial kernel;
3. RBF kernel.

Use the same image representation and data split as Part 1. Choose reasonable initial settings and **report** the following:

1. The specific training settings used;
2. The training accuracy and test accuracy;
3. The confusion matrix;
4. The total number of support vectors.

Present the main results in **one consolidated comparison table** for all kernels used. Then you should analyze the results, and in particular, answer the following questions:

1. Does a nonlinear kernel outperform the linear classifier?
2. Does higher training accuracy necessarily correspond to higher test accuracy?
3. Does the number of support vectors differ substantially between models?
4. What does this suggest about the complexity of the resulting decision boundary?

In your report, you should explain why the observed differences are reasonable for this dataset and feature representation.

b. Analyze the Interaction Between C and γ (20%)

For the soft-margin kernel SVM with RBF kernel $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$, investigate the influence of C and γ to the resulting SVM performance via a small, controlled set of combinations. For simplicity, you may set $C \in \{0.1, 1, 10\}$ and $\gamma \in \{0.1\gamma_0, \gamma_0, 10\gamma_0\}$, where $\gamma_0 = 1/d$ and d is the feature dimension. This results in only nine settings ((C, γ) pairs) and a large exhaustive search is not required. **Summarize** your results using an appropriate **table / heatmap**. For each configuration, consider:

1. Training accuracy and test accuracy;
2. The number or proportion of support vectors.

Based on your observations, **identify** examples corresponding approximately to:

1. An underfitting model;
2. A reasonably balanced model;
3. A potentially overfitting model.

Explain your selections using evidence from the experiments. Your discussion should connect the observations to the roles of:

- C : tolerance of margin violations / regularization;
 - γ : radius of influence for the RBF kernel.
-

c. Analyze Linear vs. Kernel SVM (10%)

Compare the predictions of your selected “optimal” linear SVM from Part 1c and the reasonably balanced RBF SVM from Part 2b. **Identify** representative examples from the following categories where available:

1. Both models correct;
2. Both models wrong;
3. Linear SVM wrong but RBF SVM correct;
4. Linear SVM correct but RBF SVM wrong.

Display the corresponding images and predictions. The main analysis should focus particularly on Categories 3 and 4. You can analyze based on the following questions:

1. What characteristics might require a nonlinear decision boundary?
2. Why might the RBF classifier succeed where the linear classifier fails?
3. Are there cases where the nonlinear classifier appears to overfit?
4. Could some failures still be caused primarily by the image representation rather than the classifier?

If one of the categories does not occur in your results, state this explicitly rather than artificially selecting an example.

d. Overall Conclusion over “Optimal” SVM Settings (10%)

Based on your own experimental evidence obtained from ALL the previous Parts, provide a **concise conclusion** addressing the following:

1. When was the soft-margin linear SVM sufficient?
2. When did the kernel SVM provide a meaningful advantage?
3. How did (C) affect the classifiers?
4. How did (γ) affect the RBF SVM?

5. What did the support vectors reveal about the dataset?
6. What appears to be the principal limitation of the final classifier: the SVM formulation, the kernel, or the image representation?

Please avoid generic statements such as: “RBF is better because it can model nonlinear boundaries.” Instead, support conclusions using specific observations from your experiments.